# Appendix III – Maine Aquatic Barrier Prioritization Tool - Data Sources – 2026

| Select Entity Names | Entity Abbreviation |
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| Maine Department of Inland Fisheries and Wildlife | IFW |
| Maine Department of Marine Resources | DMR |
| Maine Geological Survey | MGS |
| National Oceanic and Atmospheric Administration | NOAA |
| U.S. Department of Agriculture – Natural Resources Conservation Service | USDA – NRCS |
| U.S. Fish and Wildlife Service | USFWS |
| United State Geologic Survey | USGS |
| The Nature Conservancy, Maine | TNC Maine |
| Gulf of Maine Distinct Population Segment | GoM DPS |

| Data Layer Name: | Alewife Ponds |
| --- | --- |
| Data Description | Ponds with known spawning runs of alewife. TNC Maine worked with DMR to reconcile the data as shown in the Maine Stream Habitat viewer with data that DMR had provided TNC previously. DMR indicated (in a spreadsheet) which ponds should be added to the previous TNC dataset. This dataset is now the best available. |
| Data Source | DMR; TNC Maine |
| Date | 10/2024 |
| Date Note | Date listed is date of last edits by TNC based on information from DMR. |

| Data Layer Name: | Aquifers |
| --- | --- |
| Data Description | Dataset contains polygons of significant aquifers (glacial deposits that are a significant ground water resource) for Maine mapped at a scale of 1:24,000, from the MGS. Printed maps published by the MGS on USGS 7.5' quadrangle bases. Aquifer boundaries delineated and digitized by the MGS from data compiled on USGS 7.5' quadrangle bases. Aquifer polygons coded by yield expected from a properly constructed well (using the AQUIFERTYPE field or the field ATYPE = 0 area not mapped as aquifer; ATYPE = 1 10-50 gallons-per-minute; ATYPE = 2 greater than 50 gallons-per-minute; ATYPE = 3 island of non-aquifer material within an area mapped as aquifer). |

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| **Data Source** | MGS |
| **Date** | April 22, 2020 |
| **Date Note** | Date listed is date of last update by MGS |

| **Data Layer Name:** | **Atlantic Salmon Critical Habitat** |
|---|---|
| **Data Description** | Critical habitat for Atlantic salmon (GoM DPS) includes the stream channels within the designated stream reaches and includes a lateral extent as defined by the ordinary high-water line (33 CFR 329.11). In areas where the ordinary high-water line has not been defined, the lateral extent will be defined by the bankfull elevation. **Bankfull elevation** is the level at which water begins to leave the channel and move into the floodplain and is reached at a discharge which generally has a recurrence interval of 1 to 2 years on an annual flood series. Critical habitat in estuaries is defined by the perimeter of the water body as displayed on standard 1:24,000 scale topographic maps or the elevation of extreme high water, whichever is greater. Watershed boundaries provide a standard level of detail that is an acceptable unit to describe salmon habitat, and the watershed level 5 (HUC 10s) are nationally recognized by the USGS to spatially relate any dataset that contains these codes. |
| **Data Source** | USDA-NRCS; NOAA |
| **Date** | July 19, 2023 |
| **Date Note** | Date listed is date of last edits. |

| **Data Layer Name:** | **Atlantic Salmon Habitat Quality** |
|---|---|
| **Data Description** | Stream and River flowlines (based on 1:24K National Hydrography Dataset) rated for Atlantic salmon Habitat Quality (Low, Medium, High, Very High). Flowlines may be rated highly as spawning and rearing habitat or as important migratory rivers and streams. Data were assembled by the DMR with help from TNC Maine. |
| **Data Source** | DMR; TNC Maine |
| **Date** | 10/2024 |
| **Date Note** | Date listed is date of last edits by TNC based on information from DMR. |

| Data Layer Name: | Atlantic Salmon Habitat Recovery Units (GoM Distinct Population Segment) |
|---|---|
| Data Description | Polygon of the GoM DPS for Atlantic salmon which is divided into three sections: Merrymeeting Bay, Penobscot, and Downeast salmon habitat recovery units (SHRUs). Data sourced from DMR Open Data |
| Data Source | USFWS, DMR |
| Date | October 4, 2021 |
| Date Note | Date listed is date of last edit. |

| Data Layer Name: | Atlantic Salmon Modeled Habitat Units |
|---|---|
| Data Description | GIS-based habitat model that predicts the amount of accessible Atlantic salmon rearing habitat throughout the GoM DPS. The model was developed using data from habitat surveys conducted in the Machias, Sheepscot, Dennys, Sandy, Piscataquis, Mattawamkeag, and Soudabscook Rivers. The model uses reach slope derived from contour and digital elevation model (DEM) datasets, cumulative drainage area, and physiographic province to predict the total amount of rearing habitat within a reach. The variables included in the model explain 73% of the variation in rearing habitat. |
| Data Source | USFWS Gulf of Maine Coastal Program; NOAA |
| Date | April 2012 |
| Date Note | Date model completed. |

| Data Layer Name: | Atlantic Salmon State/Federal Active Management Areas |
|---|---|
| Data Description | Similar to Atlantic salmon Habitat Quality. Dataset represents ranked management priority within State and Federal Active Management Areas – that is, within watersheds classified as Atlantic salmon Critical Habitat. Stream and River flowlines (based on 1:24K National Hydrography Dataset) rated for Atlantic salmon Habitat Quality (Low, Medium, High, Very High). Flowlines may be rated highly as spawning and rearing habitat or as important migratory rivers and streams. |
| Data Source | DMR with help from TNC Maine |
| Date | 10/2024 |
| Date Note | Date listed is date of last edits by TNC based on information from DMR. |

| Data Layer Name: | Atlantic Salmon Surveyed Spawning and Rearing Habitat |
|---|---|
| Data Description | This coverage was developed from field surveys conducted on the mainstem and selected tributaries of the Aroostook, Dennys, Ducktap, East Machias, Kennebec, Machias, Passagassawakeag, Penobscot, Pleasant, Presumpscot, Saco, Sheepscot, St. George, Tunk, and Union Rivers in Maine by staff of the DMR - Division of Sea-Run Fisheries and Habitat. These surveys were conducted to identify important Atlantic salmon habitat including spawning and rearing areas. The majority of the survey data was collected using Trimble Pro, Pro-XL and GeoExplorer3 receivers and survey files were differentially corrected to provide 2-5 meter accuracy. Surveys for some reaches were collected with minimal or no GPS control points and the attributes were overlaid on a stream centerline created using either a GPS-acquired line, a line derived from MEGIS/USGS 1:24,000 hydrography data, or a line drawn as a centerline based on MEGIS digital orthophotography. The dataset includes information on habitat categories and areas, and an indication of spawning and rearing potential. This data is referred to as Level 3, or detailed habitat survey data, to be contrasted with the Level 2 habitat data which contains the most detailed data for individual habitat units. |
| Data Source | DMR |
| Date | March 29, 2024 |
| Date Note | Date of last update by DMR |

| Data Layer Name: | Brook Trout Habitat Rating (MDIFW 2024) |
|---|---|
| Data Description | This database is a synthesis of expert recommendations from fisheries biologists from both IFW Division of Fisheries & Hatcheries and DMR Division of Sea-Run Fisheries and Ecosystems. The purpose is to rank priority stream networks and individual reaches through each of the state's sub watersheds for the conservation and restoration of habitat for native fishes. The primary intended use of the Wild Brook Trout Habitat Rating effort is to identify areas for a variety of conservation actions, such as land protection strategies, stream connectivity enhancement or instream habitat improvement. Reaches were classified into one of 5 eligible categories: Very High Value, High Value, Moderate Value, Low Value, or NOT Connect. The NOT Connect category identifies stream reaches to maintain barriers to keep invasive exotic fish from accessing native species habitats in other priority waters. To date, this database has focused on identifying the best current and potential future habitat |

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| | for Eastern brook trout, landlocked Atlantic salmon, Atlantic sea-run salmon, and sea-run rainbow smelt. |
| **Data Source** | IFW, with assistance from TNC Maine |
| **Date** | September 17, 2024 |
| **Date Note** | Date listed is date of last edits. A downloaded version of the 2024 data was used for the analysis and displayed in the 'Layers' tab because the analysis is completed before the results are uploaded to the tool and cannot be interactive. This way, users can visualize what data is underpinning the results. |

| **Data Layer Name:** | **Calcareous/Moderately Calcareous Bedrock** |
|---|---|
| **Data Description** | Regional dataset of calcareous and moderately calcareous bedrock from TNC's Ecological Land Unit raster. |
| **Data Source** | The Nature Conservancy, Ecological Land Units 2006 as referenced in: (https://www.conservationgateway.org/ConservationPlanning/SettingPriorities/EcoregionalReports/Documents/NAP_sections070131.pdf) |
| **Date** | 2006 |
| **Date Note** | Date listed is date of Northern Appalachians/Acadian Ecoregional Conservation Assessment that utilized the Ecological Land Units |

| **Data Layer Name:** | **Coarse Sediments** |
|---|---|
| **Data Description** | Regional dataset of coarse surficial sediments from TNC's Ecological Land Unit raster. |
| **Data Source** | The Nature Conservancy, Ecological Land Units 2006 as referenced in: (https://www.conservationgateway.org/ConservationPlanning/SettingPriorities/EcoregionalReports/Documents/NAP_sections070131.pdf) |
| **Date** | 2006 |
| **Date Note** | Date listed is date of Northern Appalachians/Acadian Ecoregional Conservation Assessment that utilized the Ecological Land Units |

| Data Layer Name: | Conservation Lands/Secured Lands |
|---|---|
| Data Description | Conservation Lands / Secured Lands maintained by TNC Maine. This dataset is, in many ways, similar to the Maine Office of GIS Conservation Lands Dataset but contains additional information on projects that TNC has been involved in. In some cases, it is more up-to-date than the Office of GIS dataset, but may be missing some smaller conservation lands parcels that are present in the Office of GIS dataset. Conservation Lands included in the current analysis included those with gap status 1,2,3, and 39 – based on TNC's modified version of USGS's Gap Status Codes. |
| Data Source | TNC Maine |
| Date | 2/6/2025 |
| Date Note | Date listed is date of last update by TNC Maine. |

| Data Layer Name: | Dams |
|---|---|
| Data Description | Point dataset of dam locations in Maine maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Current dataset is a modified version of dams point feature class in the 2022 Maine stream barrier database. TNC Maine conducted an extensive QA/QC and editing effort of the 2022 dataset in late 2024 using American Rivers Dam Removal Database 2024, Atlantic States Marine Fisheries Commission Fish Passage database from DMR (primarily used for confirming and editing fish passage structure information), and Maine Department of Environmental Protection Dam Dataset 2024. The resultant dataset contains approximately 167 edits to the 2022 dam points including some additions of dams that are currently barriers and dams that were removed or modified in the past that had been removed from past versions of the dataset. |
| Data Source | TNC Maine |
| Date | 01/10/2025 |
| Date Note | Date listed is date of last publication of updated dataset from Alex Abbott after incorporation of TNC 2024 edits. |

| Data Layer Name: | Eastern Brook Trout Joint Venture Wild Trout Patches |
|---|---|
| Data Description | Allopatric and Sympatric brook trout patches. A "patch" is defined as a group of contiguous catchments occupied by wild trout (Hudy et. al. 2013). Patches are not connected physically (i.e., they are separated by a dam, unoccupied warm water habitat, downstream invasive species, etc.) and are |

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| | generally assumed to be genetically isolated. Allopatric refers to eastern brook trout only in a catchment. Sympatric refers to brook trout co-residing and competing with brown and rainbow trout. |
| **Data Source** | Eastern Brook Trout Joint Venture |
| **Date** | February 2025 |
| **Date Note** | Date listed is date TNC received latest (2025) data release from IFW |

| **Data Layer Name:** | **EcoSheds Catchments** |
|---|---|
| **Data Description** | The EcoSheds brook trout occupancy model predicts probability of brook trout occupancy for catchments smaller than 200 sq. km in the northeastern U.S. from Maine to Virginia under various future temperature scenarios. |
| **Data Source** | USGS/EcoSheds |
| **Date** | June 11, 2024 |
| **Date Note** | Date listed is date TNC downloaded latest catchments and .csv model results from ecosheds. |

| **Data Layer Name:** | **Fish Species of Management Concern / Invasive Fish Species** |
|---|---|
| **Data Description** | River and lakes with suspected and confirmed occurrences of species of management concern including black crappie, largemouth bass, muskellunge, northern pike, smallmouth bass and walleye. Data was used in the form of one feature class for each species. |
| **Data Source** | IFW |
| **Date** | 2018 (Approximate) |
| **Date Note** | Date listed is approximate date last updated by IFW |

| **Data Layer Name:** | **Floodplains** |
|---|---|
| **Data Description** | 100-yr floodplain obtained from FATHOM Fluvial & Pluvial models designed to approximate the riparian zone within which various landcover and other metrics were generated |
| **Data Source** | Fathom, Inc. |
| **Date** | 2019 |
| **Date Note** | Date listed is date TNC Maine acquired data from Fathom |

| Data Layer Name: | Hydrography |
|---|---|
| Data Description | Modified version of 1:24,000 High Resolution National Hydrography Dataset (NHD) flowline dataset. Modifications made by TNC Maine and Alex Abbott, Independent Stream Restoration Specialist. Modifications include edits to construct a single-flowline dendritic network, addition of Canadian flowline data for rivers downstream of Maine, and edits to specific sections of flowlines based on field visits, aerial photography, and LiDAR-based Digital Elevation Models. |
| Data Source | USGS; TNC Maine |
| Date | 2023 |
| Date Note | Date listed is year of last known modifications made to Hydrography dataset. No updates were made in 2024/2025. |

| Data Layer Name: | Inland Wading Bird and Waterfowl Habitat |
|---|---|
| Data Description | This dataset represents Inland Waterfowl and Wading bird Habitat (IWWH), a Significant Wildlife Habitat defined under Maine's Natural Resources Protection Act (NRPA).<br><br>This dataset was developed in accordance with Maine's NRPA. Under this Act, IFW is designated as the authority for determining Significant Wildlife Habitats (SWHs). This dataset represents IWWH. Polygons with a high or moderate rating meet the Significant Wildlife Habitat definition and are protected under NRPA; low-rated polygons do not meet the definition and are not protected under NRPA. COMPREHENSIVE REVIEW OF ORGANIZED TOWNS: Boundaries and ratings of all IWWHs in Maine's organized townships were updated in 2008. These updates were based on interpretation of high-resolution aerial imagery captured between 2001 and 2007. FIELD ASSESSMENT OF LOW/MODERATE IWWHs: Questionable wetlands that scored near the low/moderate rating threshold are being field assessed to confirm or, if necessary, modify their ratings. These field assessments were conducted in MDIFW Region A (southern/western Maine) in 2014 and 2015; the updates were published in this layer on June 13, 2016. Assessments for MDIFW Region B (western/mid-coast Maine) were conducted in 2016 and 2017. Some of those updates (approximately 40 IWWHs) were published on Oct 2, 2017; the rest were published on Dec 14, 2017. Remapping for Region C began in 2018. This version (Mar 16, 2018) includes updating mapping for the towns of Amherst, Aurora, Blue Hill, Brewer, Bucksport, Clifton, Dedham, Eastbrook, Eddington, Ellsworth, Fletchers Landing, Great Pond, Holden, |

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| | Mariaville, Orland, Orrington, Osborn, Otis, Penobscot, Surry, Verona Island, and Waltham. COMPREHENSIVE REVIEW OF UNORGANIZED TOWNS: Boundaries and ratings of all IWWHs attributes in Maine's unorganized townships are being updated. This update began in 2016 and is ongoing. It is based on interpretation of high-resolution aerial imagery captured primarily between 2013 and 2016. These updates are published approximately every 3 months for the townships completed during that period (typically about 5). The last such update was published on 2017-12-14. |
| **Data Source** | IFW |
| **Date** | April 2, 2018 |
| **Date Note** | Date listed is date of publication. |

| **Data Layer Name:** | **Land Cover (NLCD 2021)** |
|---|---|
| **Data Description** | 2021 National Land Cover Database published July 31, 2023 by a consortium of federal agencies. Resolution: 30m, Coverage: Conterminous United States, Classification: 20 Classes. For this analysis, a Maine (and immediate surroundings) version was extracted and various classes were grouped for summarization for various metrics (e.g. agricultural summarizations include both hay/pasture and cultivated crops). |
| **Data Source** | Multi-Resolution Land Characteristics (MRLC) Consortium |
| **Date** | 7/31/2023 |
| **Date Note** | Date listed is date of publication. On-the-ground condition date is 2021. |

| **Data Layer Name:** | **Large Land Ownership** |
|---|---|
| **Data Description** | Large Landownerships of Maine. Originally published by JW Sewall Co., used in-part as method for road-stream crossing ownership classification. |
| **Data Source** | Originally published by JW Sewall Co. 2018, internal edits made by TNC Maine and Forest Society of Maine. |
| **Date** | 2018 |
| **Date Note** | Date listed is date of original publication. |

| Data Layer Name: | Maine Heritage Fish Waters |
|---|---|
| Data Description | These data represent the list of designated State Heritage Fish Waters. State Heritage Fish Waters are lakes and ponds that contain State Heritage Fish, identified as Eastern Brook Trout and Arctic Charr, that have never been stocked according to any reliable records or that according to reliable records have not been stocked for at least 25 years. The list of State Heritage Fish Waters may be amended by rule and is generally reviewed and updated annually. Updated:M. Lubejko on 11.26.18: merged Tim P (2362) and Mud Pond (2360) into one polygon "TIM POND (INCLUDING MUD POND)" and swapped the locations of Lambert Pond (9775) and Unnamed Pond (9371). M. Lubejko on 1.7.19: Reduced the area of 4088 (a HW) by clipping out the boundaries of 7748 (not a HW).M. Gallagher on 12/01/2019: Added Blue Pond (1468), Martin Lake (1858), Elm Pond, Little (2444), Tobey Pond (4078), Tory Hill Pond (2334). Removed Line Pond (5162), Muscalsea Pond, Little (4034), Roberts Pond (5164). M. Gallagher on 1/3/2022: Removed Butcher Lake (1322). |
| Data Source | IFW |
| Date | February 11, 2025 |
| Date Note | Date listed is date of analysis run using live feature service from Maine Office of GIS/IFW. |

| Data Layer Name: | Natural Barriers |
|---|---|
| Data Description | Point dataset of potential natural barriers to fish passage from the Maine stream barrier database maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Originally sourced from multiple datasets, many of the natural barriers were surveyed as part of the original road-stream crossing surveys conducted 2007-2019. The dataset is currently maintained as part of the Maine stream barrier database. Updates since 2019 have been sporadic. |
| Data Source | TNC Maine |
| Date | 01/10/2025 |
| Date Note | Date listed is date of last publication of updated dataset from Alex Abbott after incorporation of TNC 2024 edits. |

| Data Layer Name: | Road-Stream Crossings |
|---|---|
| Data Description | Point dataset of road-stream crossings from the Maine stream barrier database maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Original surveys were conducted 2007-2019. Updates since 2019 have been sporadic. |
| Data Source | TNC Maine |
| Date | 01/10/2025 |
| Date Note | Date listed is date of last publication of updated dataset from Alex Abbott. |

| Data Layer Name: | Sea-Run Brook Trout Streams |
|---|---|
| Data Description | Likely Habitat for Sea-Run Brook Trout (1:24,000 flowlines). In the current and previous iteration of the Aquatic Barrier Prioritization Tool, the team at TNC Maine was advised to create a new Sea-Run Brook Trout dataset using the Coastal Streams layer created by TNC Maine in combination with the IFW Brook Trout Habitat data listed above. Coastal Streams (created using the TNC method, see 'Glossary of Metrics') were selected where IFW Brook Trout Habitat ranking was High or Very High. The result was used as the Sea-Run Brook Trout Streams data layer. |
| Data Source | TNC Maine; IFW |
| Date | September 17, 2024 |
| Date Note | Date listed is date of last edits to Brook Trout Habitat data. |

| Data Layer Name: | Smelt Reaches |
|---|---|
| Data Description | In 2024 TNC worked with various DMR point datasets to create a flowline (1:24,000) dataset of known smelt habitat. Flowlines ranked as 'High' and 'Very High' were used in the current prioritization analysis. Original DMR point data description: Points symbolize documented rainbow smelt spawning sites and waterways. Sites have been identified as historical or active spawning grounds by one of three sources: DMR list of known spawning sites written in 1971; GIS file created in 1984 from information gathered by the USFWS and DMR marking probable spawning sites and waterways used for passage for all diadromous species, attribute table allowing species specific information to be selected; and surveys conducted by Maine Marine Patrol in 2005, 2007, 2008 and 2009 in which officers visited streams identified in 1971 and 1984 as well as streams where they knew active |

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| | spawning occurs. In 2005, 2007, 2008 and 2009, the presence of adults or the presence of eggs was used to confirm spawning at the location. |
| **Data Source** | DMR, TNC Maine |
| **Date** | June 2024 |
| **Date Note** | Date listed is date TNC completed translation of existing DMR point data to prioritized flowlines for current analysis. |

| **Data Layer Name:** | **Tidal Wading Bird and Waterfowl Habitat** |
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| **Data Description** | Tidal Waterfowl and Wading bird Habitat (TWWH) is a Significant Wildlife Habitat defined under Maine's Natural Resources Protection Act (NRPA).<br><br>This dataset was developed in accordance with Maine's NRPA. Under this Act, DMR is designated as the authority for determining Significant Wildlife Habitats (SWHs). This dataset represents Tidal Waterfowl and Wading bird Habitat (TWWH).<br><br>This coastal habitat that supports tidal waterfowl and wading birds is composed of four habitat components: mudflats, emergent wetlands (salt marshes), aquatic beds (eelgrass), and reefs (mussel bars). In 2016 IFW remapped the mudflat and salt marsh components from high-resolution aerial imagery and incorporated the most current mapping of eelgrass and mussel bars DMR. Rating of these habitats under NRPA is based on size of the habitat area and proximity to other tidal habitats. Contact IFW for information on a specific TWWH polygon. |
| **Data Source** | IFW |
| **Date** | April 2, 2018 |
| **Date Note** | Date listed is date of publication. |

| **Data Layer Name:** | **Tidal Waters / Sea-level Rise Scenarios** |
|---|---|
| **Data Description** | This dataset approximates the potential inland extent of inundation from several scenarios (1.2, 1.6, 3.9, 6.1, 8.8 and 10.9 feet) of sea level rise or storm surge along the Maine coastline on top of the Highest Astronomical Tide. That Highest Astronomical Tide layer displays the maximum predicted astronomical high tide for the current National Tidal Datum Epoch (1983-2001). The sea level rise |

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| | scenarios were developed by using available long-term sea level rise data from Portland, Bar Harbor, and Eastport tide gauges and the US Army Corps of Engineers Sea-Level Change Curve Calculator and sea level rise scenarios established by NOAA et al. (2017) prepared for the US National Climate Assessment. Scenarios include low, intermediate low, intermediate, intermediate high, high, and extreme sea level rise at the 50% confidence interval. The data were developed with a static (“bathtub”) inundation model that uses LiDAR topographic data as a base digital elevation model, and first adjusts Highest Astronomical Tide tidal predictions to take into account variability in elevation datums along the Maine coastline, and then adds the storm surge/sea level rise scenarios to that initial starting elevation. The primary purpose of these data is to help inform storm surge and sea level rise vulnerability assessments and community planning. |
| **Data Source** | MGS |
| **Date** | 2021 (2018 scenarios) |
| **Date Note** | Date listed is date of publication by MGS |

| **Data Layer Name:** | **Tribal Ownership / Trust Lands** |
|---|---|
| **Data Description** | Tribal Lands in Maine as provided and updated by The Forest Society of Maine (FSM). TNC Maine provides additional updates. |
| **Data Source** | FSM with TNC Maine edits |
| **Date** | February 2025 |
| **Date Note** | Date listed is date of last updates by FSM/TNC Maine. |